

- expansion of the basis set through the spawning procedure). Many more simulations should be performed to get accurate branching ratios, but these results demonstrate the differing propensities in the two conformers.
31. These pathways are determined by linear interpolation in internal coordinates between the structures representing the Franck-Condon point and each of the excited-state

minima. The x axis is mass-weighted distance, in order to allow comparison of the two possible paths for each conformer.

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Figs. S1 to S4

Table S1

Movies S1 and S2

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A Cytochrome c Oxidase Model Catalyzes Oxygen to Water Reduction Under Rate-Limiting Electron Flux

James P. Collman,* Neal K. Devaraj, Richard A. Decréau, Ying Yang, Yi-Long Yan, Wataru Ebina, Todd A. Eberspacher, Christopher E. D. Chidsey*

We studied the selectivity of a functional model of cytochrome c oxidase's active site that mimics the coordination environment and relative locations of Fe_{a_3} , Cu_{B} , and Tyr^{244} . To control electron flux, we covalently attached this model and analogs lacking copper and phenol onto self-assembled monolayer-coated gold electrodes. When the electron transfer rate was made rate limiting, both copper and phenol were required to enhance selective reduction of oxygen to water. This finding supports the hypothesis that, during steady-state turnover, the primary role of these redox centers is to rapidly provide all the electrons needed to reduce oxygen by four electrons, thus preventing the release of toxic partially reduced oxygen species.

During the final stage of respiration, cytochrome c oxidase (CcO) catalyzes the reduction of O_2 to H_2O to power its proton pump activity. This four-electron reduction must take place without releasing toxic partially reduced oxygen species (PROS).

Although CcO has been extensively studied in single-turnover experiments, the mechanism by which the enzyme reduces O_2 under steady-state conditions, where the electron flux is rate limiting, is poorly understood (1, 2). Specifically, the role of Cu_{B} and a posttranslationally modified tyrosine residue (Tyr^{244}) have been frequent

subjects of discussion. It has been assumed that, during turnover, Cu_{B} and Tyr^{244} each deliver an electron to bound oxygen in the active site, in addition to the two electrons from the Fe in heme $_{\text{a}_3}$. This rapid intramolecular reduction in a hydrophobic site should reduce the lifetimes of potential PROS-releasing intermediates (3). The oxidized active site is then thought to be slowly recharged by ferrous cytochrome c (cyt c) (4, 5) such that O_2 only binds when both Cu_{B} and Fe_{a_3} have been reduced.

This scenario has proved challenging to test in the native enzyme, because of its inherent complexity, and in site-specific mutants, because they have been found to be inactive (6–8). We previously developed functional models of the CcO active site (9–12) and showed that a model that contains two redox sites, an iron heme, and

a distal copper, but lacking a phenol to mimic Tyr^{244} can manifest the selective four-electron catalytic reduction of oxygen at physiological pH and potential. However, our catalysts were adsorbed on graphite electrodes such that electron transfer was very rapid; in contrast to the enzyme, in which electrons arrive slowly from cyt c. We describe the steady-state catalytic reduction of O_2 by using synthetic CcO models that also mimic the Tyr^{244} site and are attached to self-assembled monolayer (SAM) films on gold electrodes in order to control electron flux. This strategy enabled us to examine the influence of each redox center (copper and phenol), while varying the rates of electron delivery and simultaneously measuring the formation of PROS electrochemically. We find that, under conditions where electron delivery limits the turnover rate, the presence of copper and a redox-active phenol to mimic tyrosine sharply reduces the formation of PROS. These model studies support the belief that both Cu_{B} and Tyr^{244} are required to selectively reduce oxygen by four electrons under the physiological condition of rate-limiting electron transfer from cyt c.

Our new structural and functional analog of the CcO active site (catalyst **1a**) possesses groups that mimic Fe_{a_3} , Cu_{B} , and Tyr^{244} (Fig. 1) (13). A myoglobin-like heme was fitted with a proximal imidazole, a distal copper was bound to a tris-imidazole ligand about 5 Å above the heme, and a phenol group was covalently attached to one of the copper-ligating imidazoles and directed toward the oxygen binding site on the heme (14). In single-turnover experiments, this fully reduced synthetic model, containing both copper and phenol, activated and reduced oxygen by

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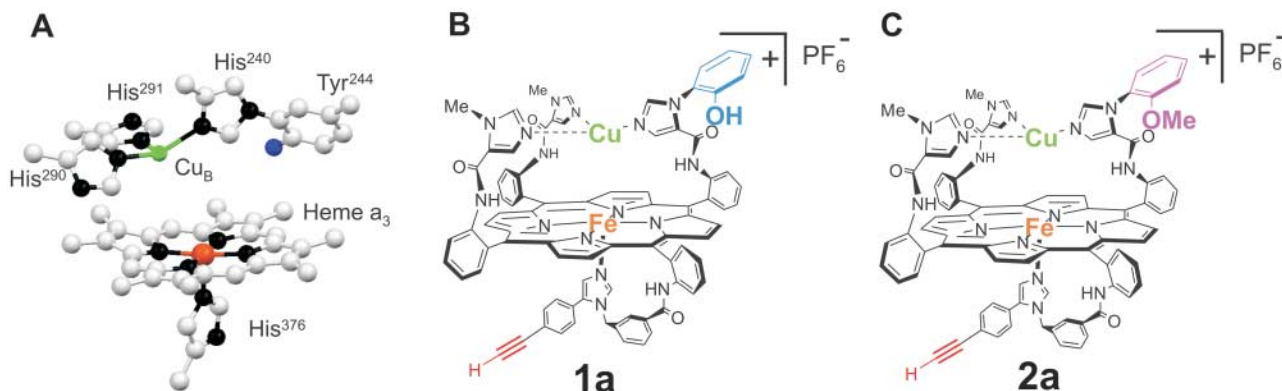
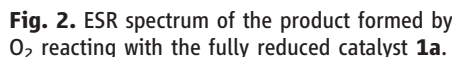


Fig. 1. (A) Crystal structure of the active site of CcO from the bovine heart (13). (B) Model **1a** reproduces the key elements of the active site of CcO. (C) Model **2a**, in which the phenol is masked as a methyl ether. Model **2a** can be treated with dilute acid to yield the iron-only model (**2b**), which is not shown.

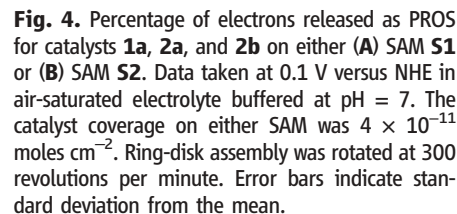
In order to control electron flux to these model compounds, we attached this model and analogs lacking copper and phenol onto mixed SAM-coated gold electrodes (15–19). Electron-transfer rates to redox groups covalently coupled to SAMs can be predictably tuned by varying the length and the degree of conjugation of the SAM (20–22). Although most functional monolayers are made by simply tethering a thiol to the redox molecule of interest, the complexity of our synthetic CcO models and the reactivity of the heme center with free thiols required an alternative method. Recently we demonstrated that azide-terminated mixed SAMs can be modified with acetylene-bearing molecules (23–25)



SAMs functionalized with CcO models were characterized by using conventional electrochemical techniques (29). The standard electron-transfer rate constant, k^0 , between the gold electrode and the iron center of our models was $6 \pm 0.1 \text{ s}^{-1}$ on SAM **S1**. In contrast, k^0 on SAM **S2** was too fast to measure with conventional electrochemical methods (28) but is at least $1 \times 10^4 \text{ s}^{-1}$. This large variation in k^0 , controlled by changing the nature of the intervening SAM, allowed us to drasti-



In order to measure the selectivity of catalysis under these two rate-limiting regimes (SAM **S1** and SAM **S2**), we used rotating ring-disk voltammetry. There are very few examples reported on the use of rotating ring-disk voltammetry with SAMs (32–34) because of the difficulty of forming reproducible SAMs on such electrodes (18). We have developed a procedure to form gold disk electrodes with the required surface properties to promote high-quality SAM formation (29). A catalyst-modified SAM-coated gold disk electrode is encircled by a PROS-detecting Pt ring electrode. The two-electrode assembly is rotated, and the gold disk is set to a potential at which oxygen reduction takes place. PROS produced during O₂ reduction are swept



from the disk to the ring, where they are electrochemically detected. For an ideal four-electron electrocatalyst, no PROS would be detected. We measured the percentage of electrons released as PROS for catalysts **1a**, **2a**, and **2b** immobilized on SAM **S1** and SAM **S2** (Fig. 4).

When attached to SAM **S2**, the incomplete CcO model **2b**, which lacks both copper and a redox-active phenol, is moderately selective at reducing O_2 to H_2O ; $89 \pm 2\%$ of the electrons are used to form H_2O . Addition of copper (catalyst **2a**) decreases the amount of peroxide detected at the ring by over 30%. Copper is known to increase the oxygen-binding affinity of both synthetic CcO models and CcO itself and to improve the selectivity of CcO models absorbed on edge-plane graphite (6, 30, 35). Interestingly, the copper- and phenol-containing catalyst **1a** on SAM **S2** shows only a slightly improved selectivity over catalyst **2a**, in which the phenol is masked. We believe this limited influence of the redox-active phenol results from the ability of the electrode to deliver an additional electron rapidly through SAM **S2**. This behavior is reminiscent of fully reduced CcO during single turnover experiments; Fe_a can rapidly ($\sim 30 \mu s$) provide an electron to a partially reduced oxygen intermediate, and a tyrosine radical is not observed (2). This result demonstrates that during turnover Tyr^{244} is not required when a fourth electron can be rapidly delivered from outside the active site.

During in vivo operation, CcO is never in the fully reduced state because electron delivery by cyt c is far slower than the rate at which internal electrons are delivered to oxygen bound

at the active site (36). Thus, studies of CcO models immobilized on SAM **S1**, where electron transfer limits turnover, have greater physiological relevance. Compared with SAM **S2**, on SAM **S1** models lacking either copper or phenol show a marked decrease in selectivity. For example, the iron-only model **2b** on SAM **S1** rapidly degrades and is likely consumed by substantial PROS formation. This is unlike the case with similar iron-only porphyrins directly absorbed on edge-plane graphite (30) but is similar to our previous work using lipid films to force slow diffusional delivery of electrons (37). When immobilized on SAM **S1**, model **2a**, which contains both iron and copper but not a redox-active phenol, is more stable than the iron-only model but still shows a marked degree of PROS leakage. This catalyst can store no more than three electrons at the active site before oxygen binding (two from iron and one from copper); therefore, one additional electron would be required from the electrode in order for oxygen to be reduced by four electrons. If the electron transfer rate between electrode and catalyst is slow, partially reduced intermediates must persist before additional electrons are delivered, and in the interim PROS can leak and degrade the catalyst (Fig. 5A).

In contrast, catalyst **1a**, equipped with both copper and phenol, is highly selective at reducing oxygen to water on SAM **S1** (Fig. 5B). Compared with the catalyst without a redox-active phenol **2a**, the complete catalyst **1a** releases over threefold less PROS. Remarkably, the selectivity of catalyst **1a** is nearly identical whether immobilized on SAM **S1** or SAM **S2**. Thus, **1a** mimics one of the most important and

fascinating features of CcO: its ability to selectively reduce oxygen by four electrons when the electron-transfer flux is rate limiting.

CcO is believed to reduce oxygen by four electrons with $>99\%$ selectivity (36). Catalyst **1a** on SAM **S1** is only about 96% selective for four-electron reduction (38). We suggest two plausible reasons for PROS formation. First, our Fe-Cu models do not require complete reduction of the metal centers in order to bind O_2 , whereas in CcO an apparent cooperativity between Fe_{a3} and Cu_B redox centers ensures that only the fully reduced active site binds O_2 (1, 36). In the present model, it is possible to form a partially reduced active site (Fe^{+2}/Cu^{+2}) that would bind O_2 . This process could lead to occasional production of PROS when electron transfer is rate limiting.

A second disparity is that these active-site models are in direct contact with water, whereas CcO is buried in a membrane. Exposure to a water interface should favor hydrolytic autooxidation, thereby increasing PROS formation (39). To test this hypothesis, we treated the SAM films with a hydrophobic surfactant (29). It has been demonstrated that such surfactant films can act as hydrophobic blocking layers and influence the physical properties of redox molecules within the SAM (40, 41). In preliminary studies, we have found that this treatment reduces the percent of electrons contributing to PROS by almost a factor of 3 (29).

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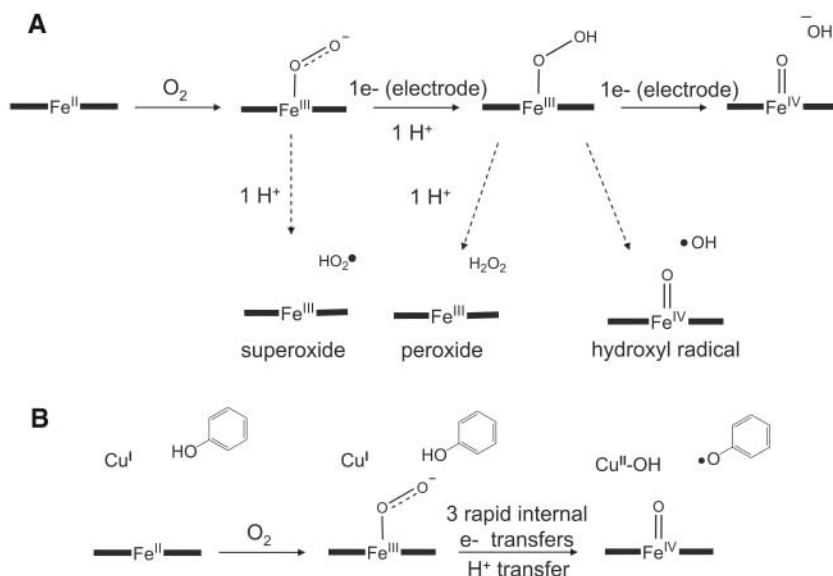


Fig. 5. Possible intermediates during the reduction of oxygen to the redox level of water. (A) A monometallic heme, such as catalyst **2b**, requires electrons to be delivered externally in order to reduce oxygen by four electrons. If those electrons are delivered slowly, the lifetimes of PROS-releasing intermediates increase. (B) Catalyst **1a** equipped with both copper and phenol can bind oxygen and rapidly reduce it to the redox level of water without requiring external electrons. PROS release is minimized.

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44. We dedicate this scientific contribution to the memory of the late H. Taube. We thank J. I. Brauman, T. D. P. Stack, C. C. Cummins, R. Boulatov, and P. H. Dinolfo for helpful discussions. N.K.D., R.A.D., and W.E. acknowledge a Stanford graduate fellowship, a Lavoisier fellowship, and a Dreyfus undergraduate fellowship, respectively. This material has its basis in work supported by the NIH under grant GM-17880-35.

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Materials and Methods

Figs. S1 to S4

References

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Thermoelectricity in Molecular Junctions

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By trapping molecules between two gold electrodes with a temperature difference across them, the junction Seebeck coefficients of 1,4-benzenedithiol (BDT), 4,4'-dibenzenedithiol, and 4,4''-tribenzenedithiol in contact with gold were measured at room temperature to be $+8.7 \pm 2.1$ microvolts per kelvin ($\mu\text{V/K}$), $+12.9 \pm 2.2$ $\mu\text{V/K}$, and $+14.2 \pm 3.2$ $\mu\text{V/K}$, respectively (where the error is the full width half maximum of the statistical distributions). The positive sign unambiguously indicates p-type (hole) conduction in these heterojunctions, whereas the Au Fermi level position for Au-BDT-Au junctions was identified to be 1.2 eV above the highest occupied molecular orbital level of BDT. The ability to study thermoelectricity in molecular junctions provides the opportunity to address these fundamental unanswered questions about their electronic structure and to begin exploring molecular thermoelectric energy conversion.

Study of charge transport in molecules is of fundamental interest, with potential applications in molecular electronics (1) and energy-conversion devices (2, 3). Current-voltage (I-V) characteristics of single molecules have been extensively investigated by trapping a single molecule in break junctions formed by mechanical strain (4), electromigration (5), and scanning tunneling microscopes (6). Although such measurements have provided substantial insight into charge transport through molecular junctions, critical aspects about the electronic structure cannot be uniquely obtained

by I-V characteristics alone. For example, whether molecular junctions are p-type or n-type—i.e., whether the position of the Fermi level, E_F , of the metal contacts is closer to the highest occupied molecular orbital (HOMO) or to the lowest unoccupied molecular orbital (LUMO)—generally remains unknown because of uncertainties in the microscopic details of the contacts (7–10). A specific example of this is benzenedithiol (BDT); in this case, although the electrical conductance has been studied extensively both experimentally (4, 11) and theoretically (7, 12–15), some groups suggest that the E_F of the electrodes lies close to the HOMO level (7, 13, 15) and other groups contend that E_F lies near the LUMO level (12, 14). Although sweeping the gate bias in single-molecule transistors could potentially yield this information, Coulombic interactions caused by charge accumulation on the molecule could perturb the electronic structure (16, 17).

It has been suggested that the sign of the Seebeck coefficient, S , of molecular junctions can indicate the sign of the charge carrier and the relative position of E_F with respect to the HOMO or LUMO levels (8). Indeed, thermo-

power measurements that use a scanning probe microscope have yielded nanoscale spatial distributions of electron and hole concentrations in inorganic semiconductors (18) and have led to chemical potential microscopy at the atomic scale (19, 20). Here, we report an alternative approach: We measured S for molecular junctions formed by trapping molecules between gold electrodes, and then we measured the voltage generated across them when a temperature bias was imposed across the junction.

In general, S is associated with bulk materials and is obtained by measuring the voltage difference created across a material in response to an applied temperature differential. In such bulk materials, charge transport is diffusive in nature. The concept of an effective S is also valid for junctions where the transport may be ballistic. For such junctions, however, a more general form of the Seebeck coefficient is needed and is given as

$$S = \frac{1}{eT} \frac{\int_0^\infty \sigma(E)(E - E_F)dE}{\int_0^\infty \sigma(E)dE} \quad (1)$$

where $\sigma(E)$ is the energy-dependent differential electrical conductivity, E_F is the Fermi level (or more accurately, the chemical potential), e is the charge of an electron, and T is the absolute temperature; the denominator in Eq. 1 is the electrical conductivity, σ . As Eq. 1 suggests, S reflects the asymmetry in $\sigma(E)$ with respect to E_F . In bulk materials, this asymmetry results from energy-dependent carrier scattering or the asymmetry in the density of states. For ballistic transport, the asymmetry can be created by a potential barrier at a junction, such as that created between E_F of a metal and the HOMO or LUMO level of a molecule. Here, S is not an intrinsic property of a material, but that of the heterojunction. Hence, we call it a junction Seebeck coefficient, S_{junction} . Because S_{junction}

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A Cytochrome c Oxidase Model Catalyzes Oxygen to Water Reduction Under Rate-Limiting Electron Flux

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